

LOGIC No: N/A (control loop only)	DESCRIPTION: SOLVENT HEATER EA-5601 - TEMPERATURE CONTROL (no dedicated ESD trip)	SIL: N/A	TAG: EA-5601 FLOC: TJC-LLD-5600-01
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CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 MODULATE STEAM VALVE TV-5602	EFF-2 ANNUNCIATE HIGH dP / FOULING ALARM	EFF-3 MECHANICAL RELIEF VIA PSV-5607
C1	Outlet temperature control	TIC-5602	SP 95 degC	control	X		
A1	Tube-side dP HIGH (fouling)	PDAH-5605	> 0.7 bar	alarm		X	
R1	Tube-side overpressure relief	PSV-5607	set 16 barg	mech			X

Legend: X = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	MODULATE STEAM VALVE TV-5602
EFF-2	ANNUNCIATE HIGH dP / FOULING ALARM
EFF-3	MECHANICAL RELIEF VIA PSV-5607

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Solvent flow established (from GA-5610)	FSL upstream
2	Condensate trap draining (TI-5604 < steam T)	TI-5604
=>	ALL permissives TRUE (AND) => START EA-5601 ENABLED	RUN

NOTES: 1. Trip set points are DUMMY training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID TJC-LLD-PID-5601 for instrument loops and the Datasheet for design limits of EA-5601.